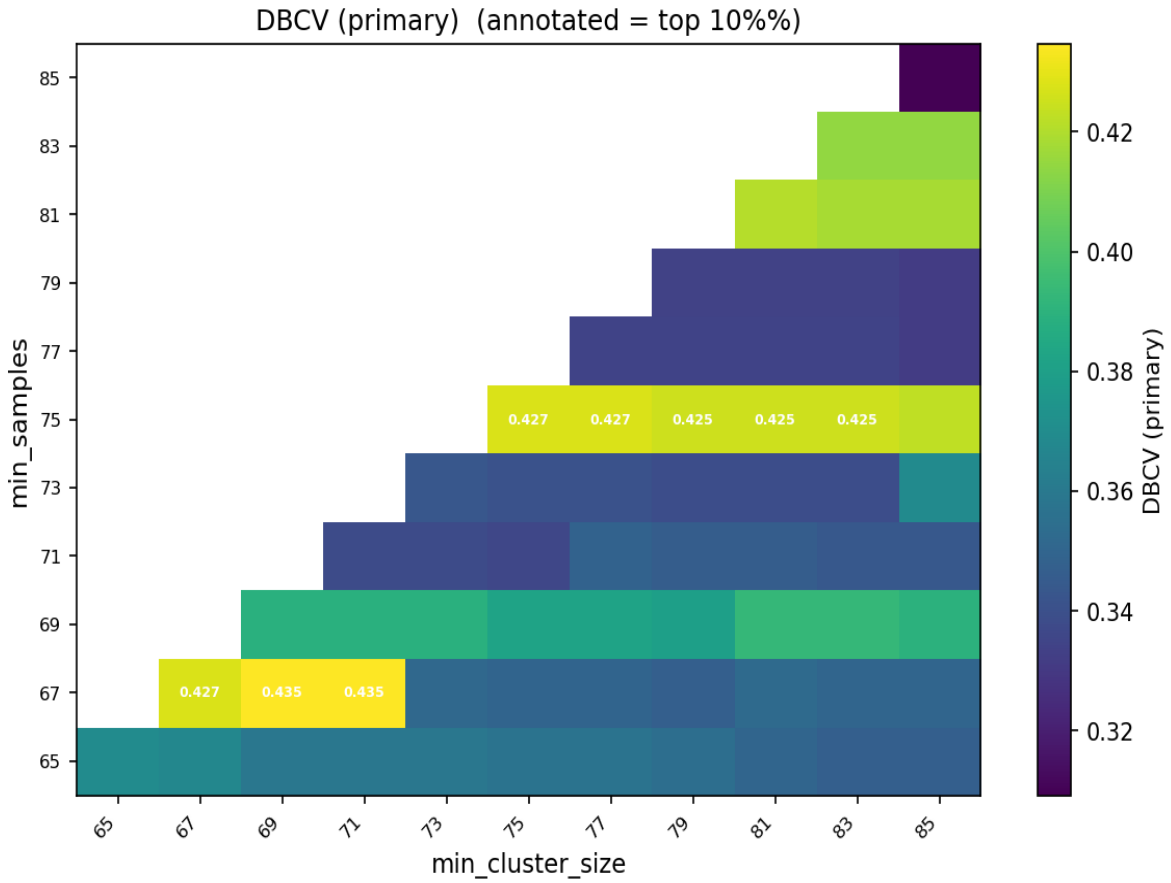
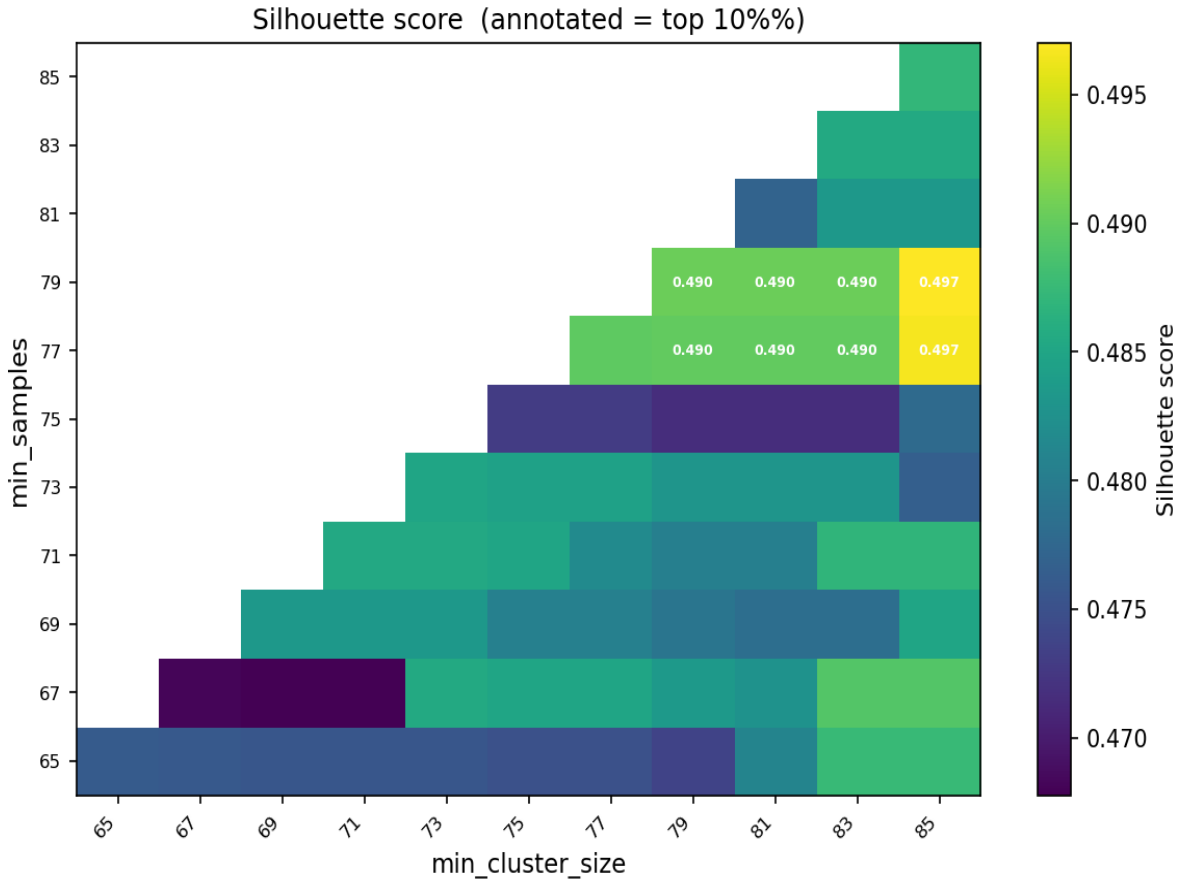
FragTree — Fine Grid Search Report

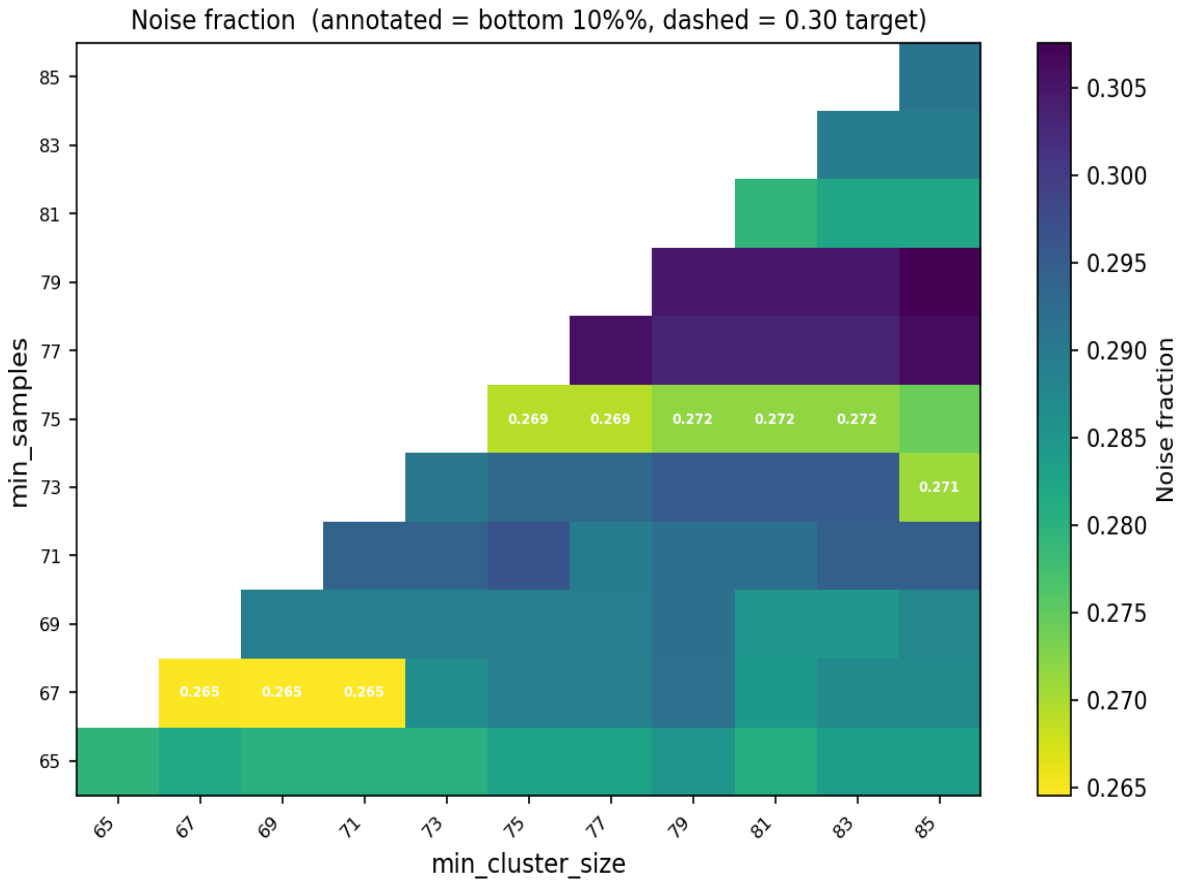
Generated: 2026-04-10 12:27:48 | n_molecules: 30,494

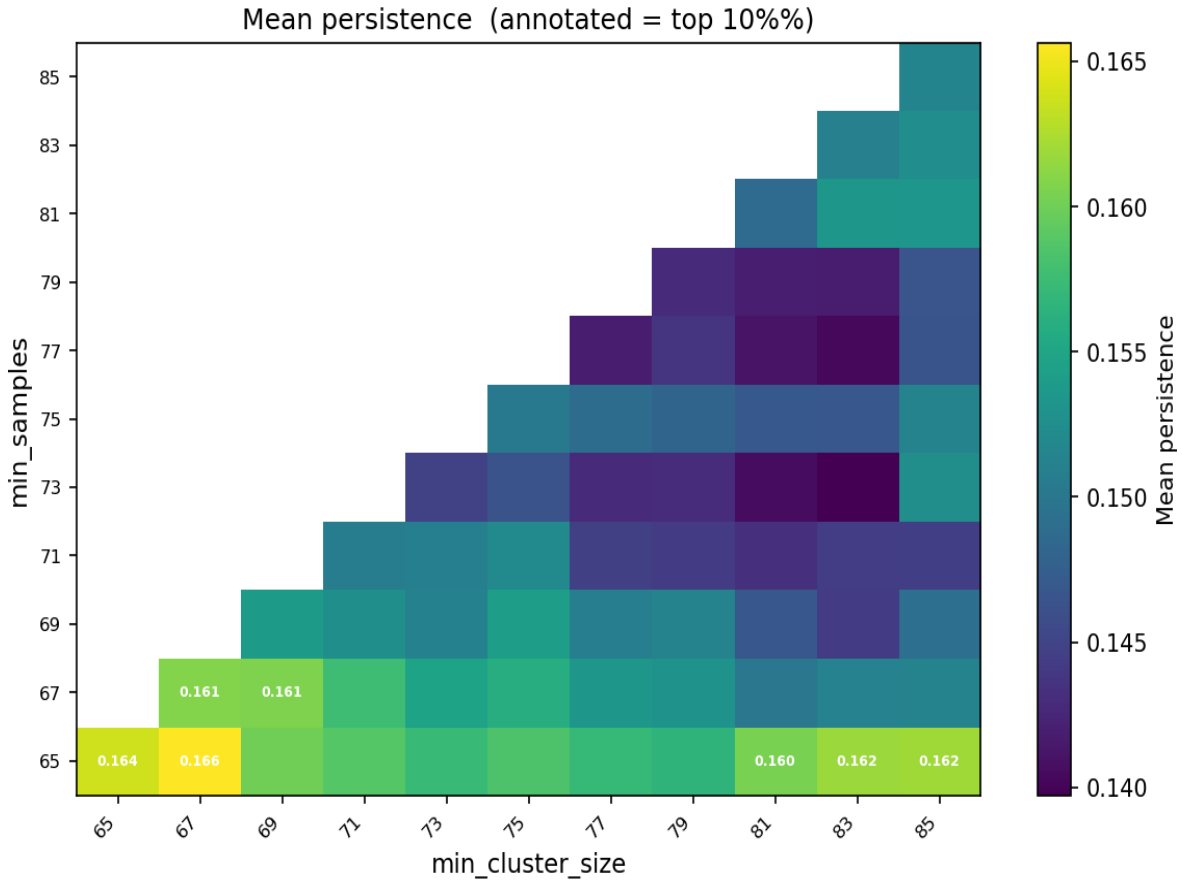
Parameter Sweep Heatmaps

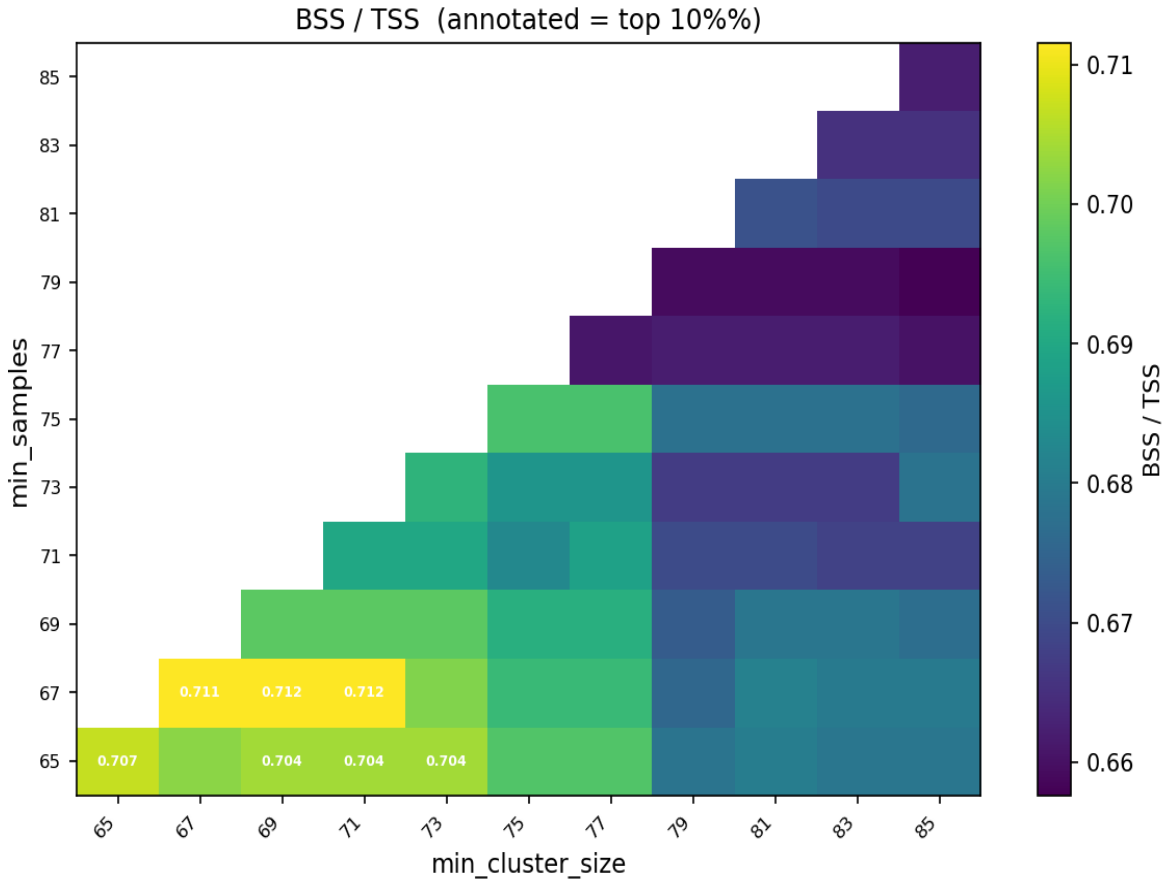
Each cell shows the metric value for a (min_cluster_size, min_samples) pair. Maximize DBCV, silhouette, mean persistence, and BSS/TSS. Minimize noise fraction (keep below 0.30). Cells with values in the top 10% are annotated with their score.




Silhouette score (annotated = top 10%%)
min_samples
min_cluster_size
Silhouette score
85
83
81
79
77
75
73
71
69
67
65
0.490
0.490
0.490
0.497
0.490
0.490
0.490
0.497
0.495
0.490
0.485
0.480
0.475
0.470


Noise fraction (annotated = bottom 10%%, dashed = 0.30 target)
min_samples
min_cluster_size
Noise fraction
85
83
81
79
77
75
73
71
69
67
65
0.269
0.269
0.272
0.272
0.272
0.271
0.265
0.265
0.265
0.305
0.300
0.295
0.290
0.285
0.280
0.275
0.270
0.265


Mean persistence (annotated = top 10%%)
min_samples
min_cluster_size
Mean persistence
85
83
81
79
77
75
73
71
69
67
65
0.164
0.166
0.161
0.161
0.160
0.162
0.162
0.165
0.160
0.155
0.150
0.145
0.140


BSS / TSS (annotated = top 10%%)
min_samples
min_cluster_size
BSS / TSS
85
83
81
79
77
75
73
71
69
67
65
0.707
0.711
0.712
0.712
0.704
0.704
0.704
0.71
0.70
0.69
0.68
0.67
0.66

Top-10% Summary Table

A parameter pair is included if it falls in the top 10% of ANY metric (or bottom 10% for noise fraction, where lower is better). Sorted by DBCV descending.

cs	ms	DBCV	Silhouette	Noise frac	Mean persistence	BSS/TSS	N clusters
69	67	0.4348	0.4678	0.265	0.1606	0.7116	49
71	67	0.4348	0.4678	0.265	0.1576	0.7116	49
67	67	0.4275	0.4682	0.265	0.1609	0.7114	50
75	75	0.4275	0.4730	0.269	0.1502	0.6962	46
77	75	0.4275	0.4730	0.269	0.1489	0.6962	46
79	75	0.4252	0.4716	0.272	0.1481	0.6777	45
81	75	0.4252	0.4716	0.272	0.1469	0.6777	45
83	75	0.4252	0.4716	0.272	0.1467	0.6777	45
65	65	0.3696	0.4762	0.280	0.1638	0.7069	54
85	73	0.3694	0.4766	0.271	0.1526	0.6782	44
67	65	0.3675	0.4759	0.282	0.1656	0.7023	53
69	65	0.3592	0.4756	0.280	0.1600	0.7040	51
71	65	0.3592	0.4756	0.280	0.1589	0.7040	51
73	65	0.3592	0.4756	0.280	0.1573	0.7040	51
81	65	0.3500	0.4810	0.281	0.1604	0.6805	48
83	65	0.3474	0.4875	0.284	0.1618	0.6789	47
85	65	0.3474	0.4875	0.284	0.1619	0.6789	47
79	77	0.3342	0.4900	0.303	0.1438	0.6619	47
81	77	0.3342	0.4900	0.303	0.1411	0.6619	47
83	77	0.3342	0.4900	0.303	0.1403	0.6619	47
79	79	0.3341	0.4904	0.305	0.1428	0.6593	47
81	79	0.3341	0.4904	0.305	0.1419	0.6593	47
83	79	0.3341	0.4904	0.305	0.1417	0.6593	47
85	77	0.3316	0.4966	0.306	0.1464	0.6602	46
85	79	0.3314	0.4970	0.308	0.1465	0.6576	46